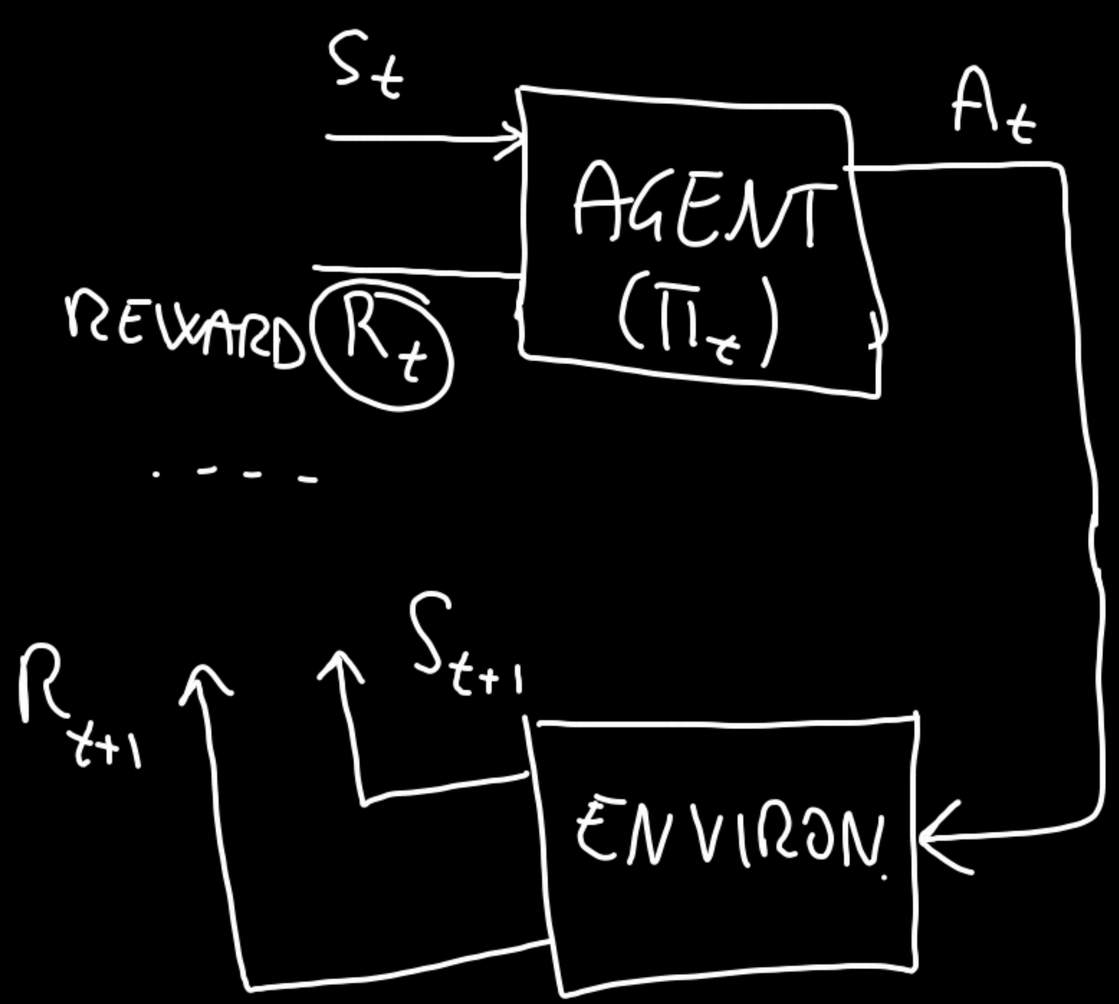
23/10/26 - RL, INTRODUCTION, VALUE FUNCTIONS



EPISODIC TASKS: TASKS THAT TERMINATE AT SOME POINT.

CONTINUING TASKS (OR NON-EPISODIC). GO ON FOREVER.

GOAL: LEARN OPTIMAL POLICIES: POLICIES THAT MAXIMIZE TOTAL RETURN ($\sum_t$ REWARD)

POLICY:

DET: $\pi : S \longrightarrow A$ (S: SET OF POSSIBLE STATES. A: SET OF POSSIBLE ACTIONS).

STOC $\pi : S \times A \longrightarrow [0,1]$. PROBABILITY OF TAKING A GIVEN ACTION IN A GIVEN STATE.

$\pi(s,a) = 0.3$

VALUE FUNCTIONS ($V_\pi(s)$, $Q_\pi(s,a)$)

$V_\pi : S \longrightarrow \mathbb{R}$. $V_\pi(s)$ IS THE EXPECTED FUTURE TOTAL RETURN OBTAINED BY THE AGENT WHEN STARTING FROM STATE S, AND FOLLOWING POLICY π THEREAFTER.

VALUE ITERATION.

π_0 — POLICY INITIALIZATION.

↓

V_{π_0} — (POLICY EVALUATION: COMPUTATION OF THE VALUE FUNCTION ASSOCIATED TO A GIVEN POLICY)

↓

π_1 — (POLICY IMPROVEMENT: MODIFY π_0 (USING INFO FROM V_{π_0}) TO GET A BETTER POLICY π_1).

↓

V_{π_1}

⋮

V^* —greedy→ π^* OPTIMAL POLICY.

(BETTER POLICY)

$$\pi_1 \geq \pi_0$$

$$V_{\pi_1}(s) \geq V_{\pi_0}(s)$$

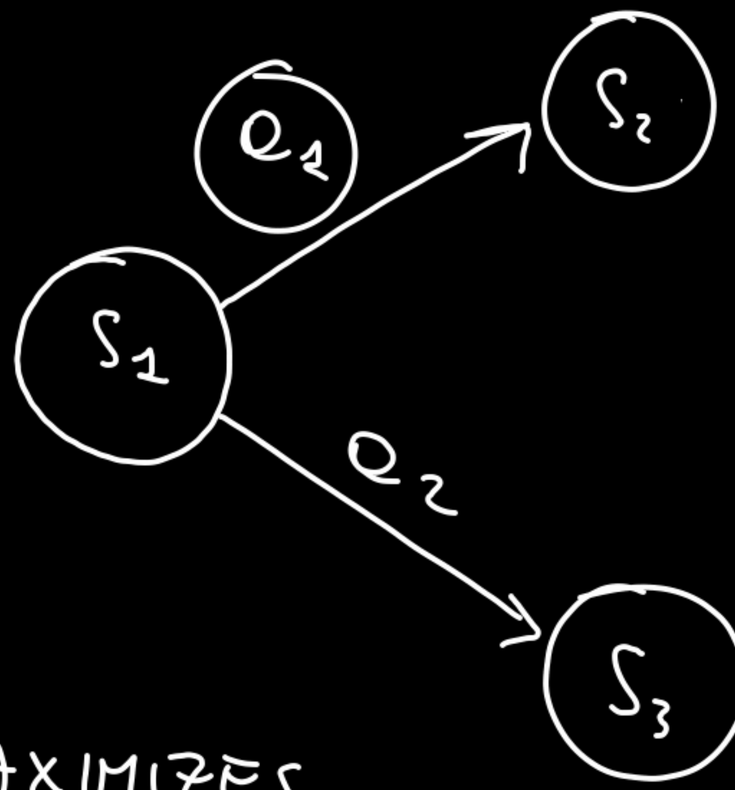
OPTIMAL VALUE FUNCTION

$V^*(s)$

$\|\|$

$V_{\pi^*}(s)$

$V^*(s_2) = 51$



$V^*(s_3) = 43$

GREEDY ACTION :

THE ACTION THAT MAXIMIZES THE VALUE OF NEXT STATE.

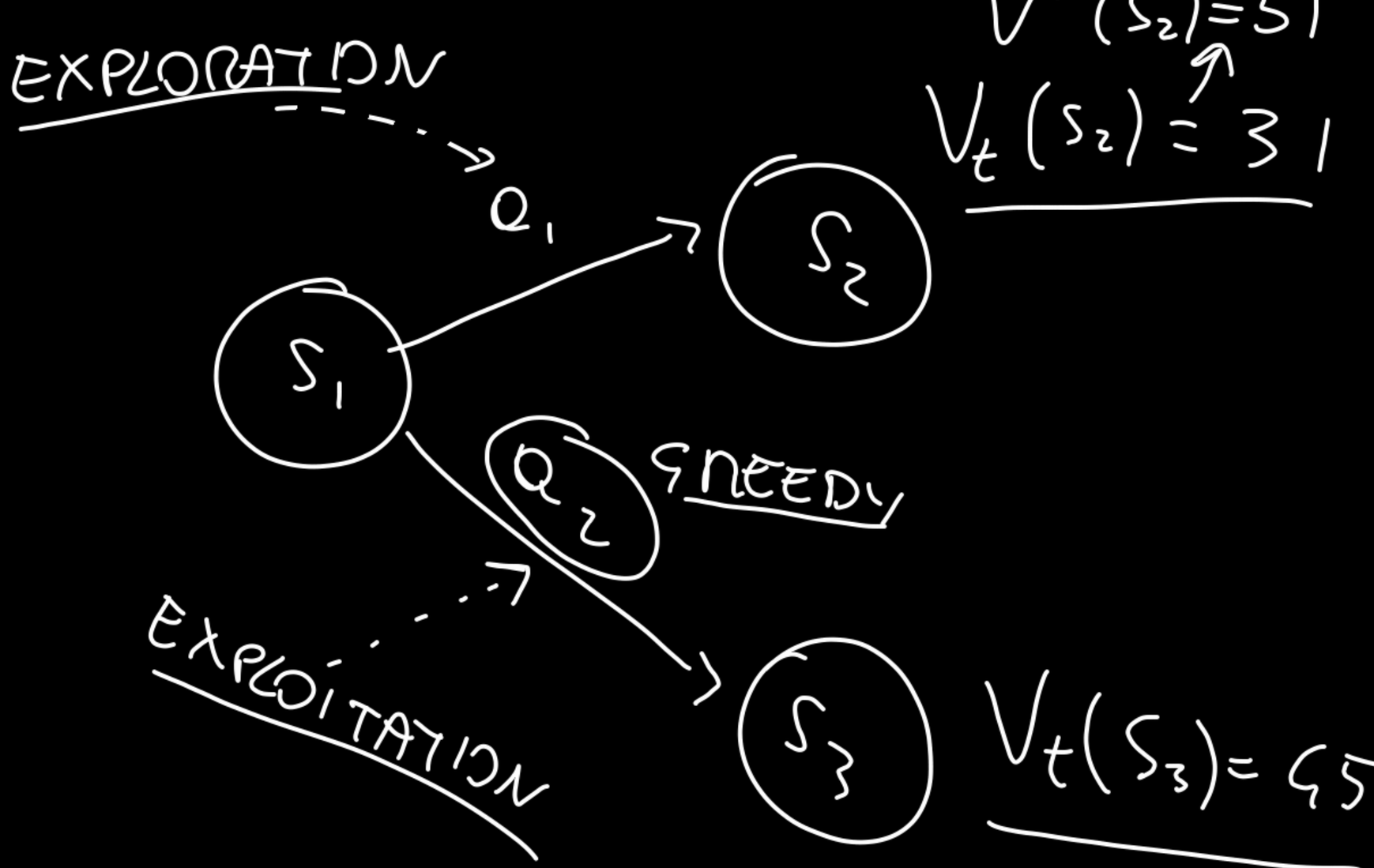
GREEDY POLICY :

A POLICY THAT ALWAYS SELECTS THE GREEDY ACTION.

1

GIVEN V^*, π^* IS THE ASSOCIATED GREEDY POLICY.

$V_t(s)$ CURRENT VALUE FUNCTION (NON OPTIMAL)



$V^*(S_2)=51$

$V_t(S_2)=31$

$V_t(S_3)=45$

GREEDY POLICY IS OPTIMAL ONLY IF WE KNOW V^*.

DURING LEARNING, WE ONLY HAVE AN APPROXIMATION OF V^* (V_t). SO THE ASSOCIATED POLICY IS NOT OPTIMAL.

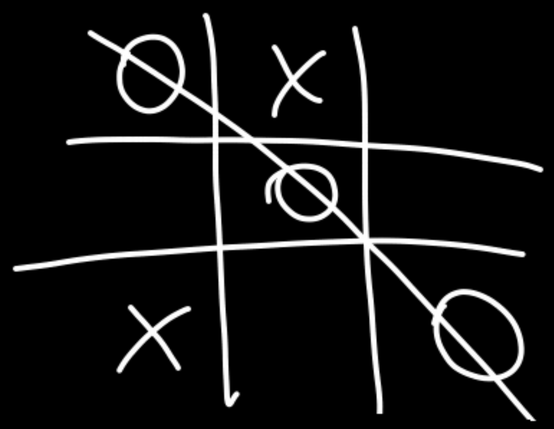
ε-GREEDY POLICY:

ε-GREEDY:

- WE TAKE THE GREEDY ACTION WITH PROBABILITY $(1-\varepsilon)$
- RANDOM ACTION WITH PROBABILITY ε.

ε_t: EXPLORATION RATE.

TWO EXAMPLES (V , Q).

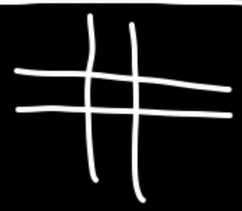
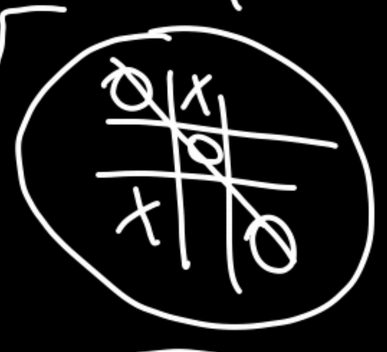


WE PLAY X

OPPONENT PLAYS O.

] LEARN HOW TO OPTIMALLY PLAY USING V.

STATE OF TRIS. (WE PLAY X).

STATE	$V_\theta(S)$
	0.5
	0
	1
	0

FINAL STATES

TABULAR REPRESENT. OF V.

V(S) IS THE PROBABILITY OF WINNING FROM S.

LET'S SEE HOW TO PLAY AND LEARN THE OPTIMAL POLICY WITH V.

BACK UP

$V(S_t) = 0.4$

S_t

OPPONENT (O)

greedy ACTION

(EXPLOIT)

(X)

0.3 0.9* 0.5 0.1 (S_{t+1})

(O)

EXPLORE

NO BACK-UP

0.3 1 0.5 (S_{t+2})

(EXPLORATION)

(X)

ε-GREEDY 0.3

greedy 1

RANDOM 0

0.3

$$V(S_t) \leftarrow V(S_t) + \alpha_t \left[V(S_{t+1}) - V(S_t) \right]$$

$0.4 + 0.1 (0.9 - 0.4)$

$$V(S_t) \leftarrow V(S_t) + \alpha_t \left[V(S_{t+1}) - V(S_t) \right]$$

TEMPORAL DIFFERENCE METHOD (TD)

OLD VALUE $\leftarrow$ OLD VALUE + STEP (NEW VALUE − OLD VALUE)

NEW VALUE → TARGET

α_t : LEARNING RATE

CONDITIONS THAT α_t MUST SATISFY TO ASSURE CONVERGENCE.

OCCUPANCY CONSTRAINT

$$W_w^{min} \leq \sum_{t : w \in W_t} c_{ik,t} \leq W_w^{max} \quad ; \forall i, w$$

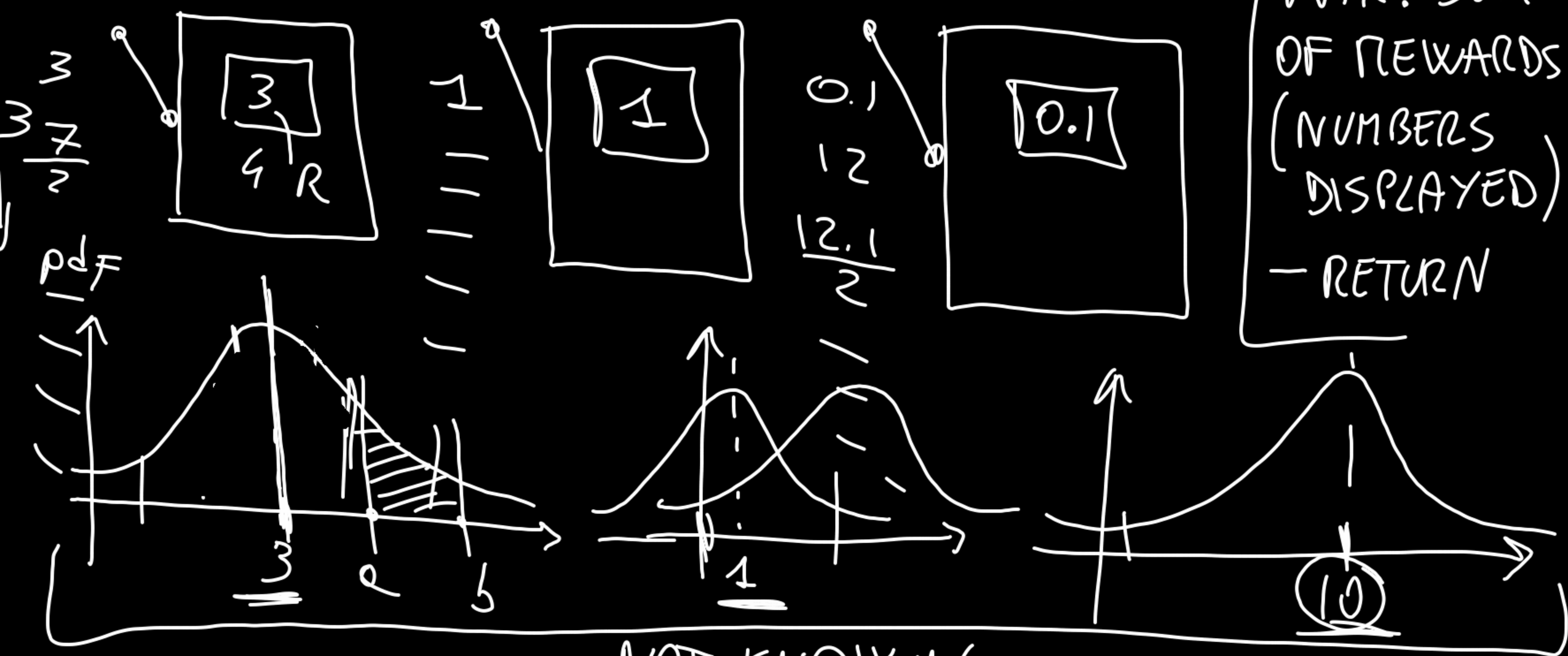
$\sum_{t : w \in W_t}$ ↑ IF TASK IS RUNNING, BUT MAYBE NOT ON w

$$W_w^{min} \leq \sum_{t : w \in W_t} \underbrace{c_{ik,t,w}}_{\text{IF } t \text{ IS RUNNING ON } w.} \leq W_w^{max} \quad ; \forall i, w$$

Q-VALUE FUNCTION

MULTI-ARMED BANDIT PROBLEM.

Q_1 Q_2 Q_3



MAX. SUM OF REWARDS (NUMBERS DISPLAYED)

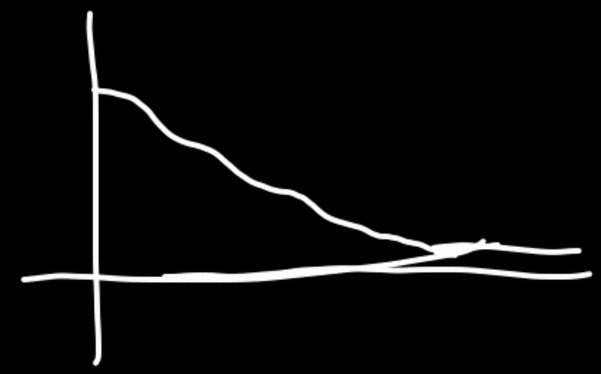
- RETURN

WE WANT TO LEARN THE OPTIMAL Q FUNCTION

$$Q^*(a) := E\{R_t \mid A_t = a\}$$

$$Q^*(a_1) = E\{R_t \mid A_t = a_1\} = 3$$

$$Q^*(a_2) = 1 \; ; \; Q^*(a_3) = 10$$



WE MUST LEARN EXPECTED VALUES.

$$Q_t(a) = \frac{\sum \text{OF REWARDS (NUMBER DISPLAYED) WHEN } a \text{ WAS SELECTED}}{\text{NUMBER OF TIMES } a \text{ WAS SELECTED.}}$$

$Q_t(a) \longrightarrow Q^*(a)$ IF NUMBER OF TIMES a WAS SELECTED $\longrightarrow \infty$.

FOCUS ON A GIVEN ACTION Q. (SAMPLE-AVERAGE)

CALL R_i THE REWARD OBTAINED WHEN Q WAS SELECTED FOR THE i-TH TIME (EX: R_3 IS THE REWARD WE GOT WHEN WE SELECTED Q FOR THE THIRD TIME.

$$Q_n(Q) = \frac{R_1 + R_2 + \cdots + R_{n-1}}{n-1}$$

ESTIMATE WE BUILD AFTER $n-1$ SELECTIONS OF Q.

ITERATIVE VERSION FOR Q_n.

· REWARD FROM LAST EXP.

$$Q_{n+1}(a) = \frac{1}{n} \cdot \sum_{i=1}^{n} R_i = \frac{1}{n} \cdot \left[\sum_{i=1}^{n-1} R_i + R_n \right] =$$

$$= \frac{1}{n} \left[\frac{(n-1)}{(n-1)} \sum_{i=1}^{n-1} R_i + R_n \right] = \frac{1}{n} \left[(n-1) Q_n + R_n \right] =$$

Q_n

$$= Q_n + \frac{1}{n} \left(R_n - Q_n \right)$$

$$\boxed{Q_{n+1} \leftarrow Q_n + \frac{1}{n} \left(R_n - Q_n \right)}$$

ACTION VALUE FUNCTION.

$Q_m(a) \rightarrow Q^*(a)$ ONLY IF $\underline{m \rightarrow \infty}$ (m IS THE NUMBER OF TIMES WE CHOOSE THAT ACTION)

$\underline{\varepsilon}$-GREEDY POLICY ASSURES THAT EVERY ACTION IS TRIED AN INFINITE NUMBER OF TIMES ($m(a) \rightarrow \infty, \forall a$) AND THIS GUARANTEES CONVERGENCE OF Q ESTIMATE TO Q^*.

ε-GREEDY POLICY

- SELECT $\underset{a}{\text{argmax}}\, Q_m(a)$ WITH PROBABILITY $(1-\varepsilon)$
- RANDOM ACTION WITH PROB. ε. ← EXPLORATION